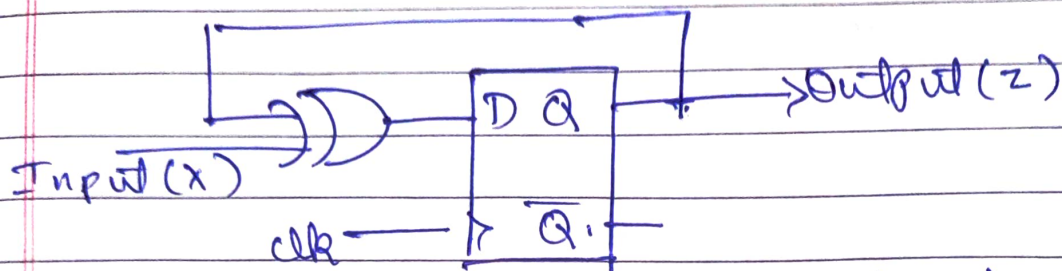


Day-4: Sequential Assignment:

DATE: / /

1.) Explain the Behavioural of the given circuit.



XOR gate inputs are

External input = x

Feedback input = Q

$$D = x \oplus Q$$

$$Q_{next} = D$$

$$Q_{next} = x \oplus Q$$

$$Z_{next} = x \oplus Z$$

State Table	
Present (Q)	Input (x) Q_{next}
0	0 0
0	1 1
1	0 1
1	1 0

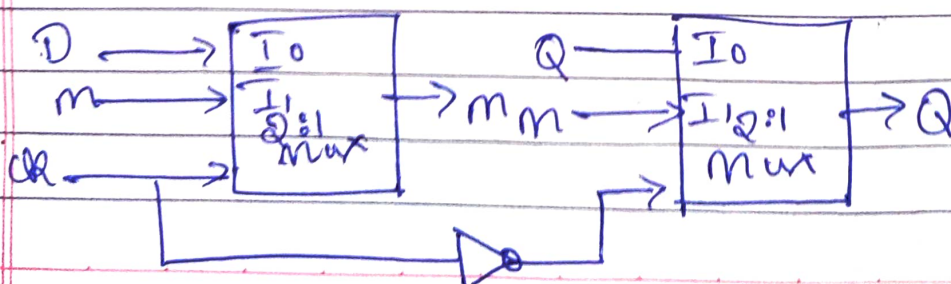
case 1: $x=0 \Rightarrow Q_{next} = Q$

case 2: $x=1 \Rightarrow Q_{next} = \bar{Q}$

The circuit behaves like a T flip flop

2.) Design a posedge triggered DFF using 2:1 mux
 only master Latch $\Rightarrow m = \bar{clk}D + clk m$
 slave latch $\Rightarrow Q = \bar{clk}m + clk Q$
 OK

Mux 1 Input	connection	Mux 2 Input	connection
I_0	D	I_0	Q (feedback)
I_1	m (feedback)	I_1	m
Select	clk	Select	clk
Output	m	Output	Q



3) Design a negative edge triggered T-FF using 2:1 mux
only characteristic equation of T-FF is $Q_{next} = T \oplus Q$

For 2:1 mux

Select line - T

$I_0 = Q$

$I_1 = \bar{Q}$

$Y = TQ + T\bar{Q}$

$Y = T \oplus Q$

$m = \bar{Q}K + Q(T \oplus K)$

$Q = \bar{Q}Q + \bar{Q}K + Qm$

Master latch $ck=1$

Input connection

I_0 m (feedback)

I_1 $T \oplus m$

Select ck

Output m

Slave latch $ck=0$

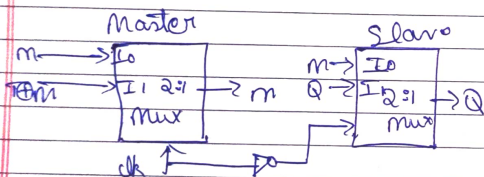
Input connection

I_0 m

I_1 Q (feedback)

Select ck

Output Q



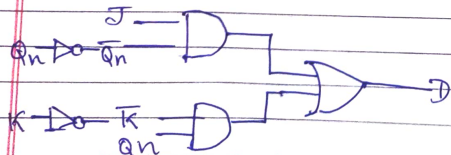
4) Convert by adding external gates

a) A D-flip flop to a JK flip flop

$Q_{n+1} = J\bar{Q}_n + KQ_n$

$Q_{n+1} = D$

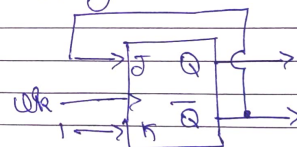
$D = J\bar{Q}_n + KQ_n$



b) A T-flip flop to a D-flip flop
 $Q_{n+1} = T \oplus Q$ $Q_{n+1} = D \Rightarrow D = T \oplus Q$
 $T = D \oplus Q$



5) In a positive edge triggered JK flip flop, we have $J=Q'$ and $K=1$ as shown in the figure below. Assume the flip flop was initially cleared and then clocked for 6 clock pulses, find the sequence at the Q output for those 6 clock cycles



$Q_{n+1} = J\bar{Q}_n + KQ_n$

$K=1 \Rightarrow K=0$ $J=Q$

$Q_{n+1} = Q \cdot \bar{Q} + 0$

$Q_{n+1} = \bar{Q}$ (Toggles on every clock pulse).

clock pulse	present Q	$J=Q'$	K	next Q
0	0	1	1	1
1	1	0	1	0
2	0	1	1	1
3	1	0	1	0
4	0	1	1	1
5	1	0	1	0
6	0	1	1	1

Sequence is 0 $\rightarrow$ 1 $\rightarrow$ 0 $\rightarrow$ 1 $\rightarrow$ 0 $\rightarrow$ 1 $\rightarrow$ 0